

Evaluating Transfer Learning for Multilabel Emotion Classification in Bantu Languages using AfriBERTa and XLM-R

Transfer Learning for Emotion Analysis in Bantu Languages

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1 Introduction

The linguistic landscape of Africa, characterized by its unparalleled diversity, presents a fertile yet underexplored frontier for Natural Language Processing research. Within this context, the Bantu language family, encompassing isiZulu, isiXhosa, and Kiswahili, shares significant linguistic similarities such as shared grammar, noun class systems, and vocabulary and represents a significant area of study, given its widespread use and the current dearth of sophisticated NLP tools.[1]

This project endeavors to address the crucial task of multilabel emotion classification, a nuanced undertaking that captures the simultaneous presence of multiple emotions within text. Recognizing the limitations posed by the scarcity of annotated datasets for these languages, this research proposes a novel approach: the systematic evaluation of transfer learning effectiveness. Specifically, it seeks to determine how effectively pre-trained multilingual models, namely AfriBERTa and XLM-R, can be fine-tuned to accurately discern and classify multiple emotions in isiZulu, isiXhosa, and Kiswahili.[2]

By pursuing this investigation, the project aims to contribute significantly to the development of culturally relevant and linguistically aware NLP applications, thereby bridging a critical gap in African language technology

2 Background and Motivation:

- The prevalence of isiZulu, isiXhosa, and Kiswahili across Southern and Eastern Africa underscores the need for robust NLP tools for these languages.
- Multilabel emotion classification is essential for understanding the nuanced emotional content of text, but the limited availability of annotated datasets for these languages poses a significant challenge.
- Transfer learning, specifically utilizing pre-trained multilingual models like AfriBERTa and XLM-R, offers a promising approach to overcome this data scarcity.
- This project aims to rigorously evaluate the effectiveness of these models in fine-tuning for multilabel emotion classification using the BRIGHTER + EthioEmo datasets, focusing on the selected Bantu languages.

3 Research Question and Objectives:

Research Question:

How effectively can pre-trained multilingual models (AfriBERTa and XLM-R) be fine-tuned for multilabel emotion classification in isiZulu, isiXhosa, and Kiswahili?

Objectives:

1. To fine-tune AfriBERTa and XLM-R models on the BRIGHTER + EthioEmo datasets for isiZulu, isiXhosa, and Kiswahili.
2. To systematically compare the performance of these models in multi-label emotion classification across the selected languages.
3. To analyze the impact of different fine-tuning strategies and hyperparameters on model performance.
4. To quantify the transfer learning effectiveness of the models.

4 Dataset and Resources:

- **BRIGHTER + EthioEmo Datasets:** Specifically isiZulu, isiXhosa, and Kiswahili subsets.
- **Pre-trained Multilingual Models:** AfriBERTa, XLM-Roberta
- **Software and Libraries:** Transformers library (Hugging Face), Pytorch or Tensorflow.
- **Supporting libraries:** scikit-learn (evaluation metrics), numpy (data manipulation and preprocessing), seaborn (visualization and analysis)

5 Methodology

5.1 Data Preparation:

Pre-process the BRIGHTER + EthioEmo datasets for the selected languages, ensuring consistent formatting, handling any language-specific nuances and partitioning of data.

- **Data Cleaning:** Remove extraneous characters and normalize punctuation.
- **Language-Specific Processing:** Apply tokenization and preprocessing techniques tailored to the morphological and syntactic properties of each language.
- **Label Formatting:** Since emotion classification is treated as a multi-label task, ensure that each sample can be tagged with multiple emotions (e.g., joy and surprise).
- **Dataset Splitting:** Partition the dataset into training, validation, and test splits while maintaining a balanced distribution of labels across splits.

5.2 Model Fine-tuning:

Fine-tune the pre-trained multilingual transformer models, AfriBERTa and XLM-R. Append a multi-label classification head on top of the pre-trained models to enable independent predictions for each emotion label. Use the validation set to tune the parameters using a grid search or random search strategy over learning rates, batch sizes, and number of epochs. Use the

training set for model fitting and the validation set for tuning and early stopping. Implement techniques like gradient clipping, learning rate warm-up, and weight decay to stabilize training and avoid overfitting and use binary cross-entropy loss suitable for multi-label classification. Apply dropout layers and monitor validation loss to mitigate overfitting.

5.3 Comparative Evaluation:

Evaluate and compare the performance of the fine-tuned models, AfriBERTa and XLM-R, across the three languages, using appropriate multi-label classification metrics as well as quantifying the training time, and computational resources required for each model.

- **F1-score:** Measures the harmonic mean of precision and recall, balancing false positives and false negatives.
 - Micro-F1: Global average across *all labels*, prioritizing frequent classes. (Good for imbalanced data.)
 - Macro-F1: Average of per-label F1 scores, treating all labels equally. (Highlights rare emotions.)
- **Precision:** The fraction of predicted positive labels that are actually correct.
 - Micro: Overall precision (correct positives / total predicted positives).
 - Macro: Average precision per label. (Ensures no emotion is overlooked.)
- **Recall:** The fraction of true positive labels correctly predicted.
 - Micro: Overall recall (correct positives / total actual positives).
 - Macro: Average recall per label. (Critical for capturing all instances of rare emotions.)
- **Hamming Loss:** The fraction of incorrectly predicted labels across all instances and labels. Lower is better. (Measures partial correctness in multilabel predictions.)
- **Subset Accuracy:** The fraction of instances where all labels are predicted exactly correctly. (Reflects nuanced emotion combinations.)
- **Computational efficiency metrics:** The resources required to train/run models.
 - Time/Memory: Training/inference speed and RAM/GPU usage.
 - Energy/Model Size: Practicality for low-resource deployment (e.g., AfriBERTa vs. XLM-R efficiency).

Macro metrics ensure rare, culturally specific emotions (e.g., nuanced terms in isiZulu) aren't drowned out by dominant labels. Hamming Loss/Subset Accuracy handle agglutinative structures and overlapping emotions. Efficiency metrics determine real-world viability in resource-constrained African NLP contexts.

5.4 Analysis of Fine-tuning Strategies:

Experiment with different fine-tuning strategies to assess their impact on performance and analyze the effect of varying hyperparameters on model accuracy and robustness.

- **Layer Freezing:** Freeze early layers of the transformer encoder to reduce training time and mitigate catastrophic forgetting, particularly in low-resource settings.

- **Gradual Unfreezing:** Begin training with only the top layers unfrozen and incrementally unfreeze lower layers across epochs, allowing for stable adaptation to new data.
- **Hyperparameter Sensitivity:** Assess the impact of different hyperparameters on performance stability and robustness across languages.

These strategies are quantitatively and qualitatively analyzed to determine best practices for fine-tuning large multilingual models in resource-constrained, multilingual emotion classification settings.

6 Expected Deliverables:

- **Proposal:** A concise research proposal outlining the approach.
- **Implementation and Experiments:** A well-documented code repository on GitHub.
- **Report:** A detailed report analyzing the performance of AfriBERTa and XLM-R, comparing their effectiveness across the selected languages and fine tuning strategies with use of BRIGHTER + EthioEmo Datasets.
- **Presentation:** A presentation summarizing the project findings and conclusions.

7 References and Further Reading

- BRIGHTER: BRIdging the Gap in Human-Annotated Textual Emotion Recognition Datasets for 28 Languages[3].
- EthioEmo: Evaluating the Capabilities of Large Language Models for Multi-label Emotion Understanding[4][5].
- AfriBERTs papers[6].
- XLM-R papers[7][8].
- Papers on transfer learning for NLP[9][10].
- Papers on multi-label classification[11].

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